

Zafar Khan

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Professional Experience

CBOE Global Markets, Full Stack Software Engineer *December 2025 - Present*

- Developing infrastructure for a greenfield operations dashboard for market data, trading ops, and analytics while migrating legacy application functionality into the new platform. (C# / React / PostgreSQL)
- Served as a core engineer on a 3-person full stack team, owning features end-to-end from database layer through UI in an Agile development environment.
- Designed and implemented role-based access control (RBAC) system for granular user permissions across the platform, along with RESTful APIs and database schema design/migrations.
- Leveraged AI-assisted development tools (Claude Code) to accelerate prototyping, code generation, and feature delivery across the stack

Wabtec Corporation, Software Engineer *June 2022 - December 2025*

- Led development/integration of a display library enabling bi-directional communication over WebSockets and OpenSSL between server and client; utilized cross-compilation between Clang and GCC in a Linux environment and integrated precompiled libraries for real-time status updates and UI rendering. (C++ / OpenSSL / Boost-ASIO, Multithreading)
- Single-handedly performed full stack development and deployment for a centralized license management system used for generating high-value license keys (\$100K+ deals). Hosted on AWS with Terraform. (C# / Python / CSHTML / MySQL, .NET6)
- Built a machine learning model (LSTM) to conduct continuous log analysis to detect anomalies and defects prior to customer reports. Model achieved 99% accuracy upon detecting known defects. (Python / PyTorch)

Wabtec Corporation, Software Engineering Intern/Co-op *May 2021- May 2022*

- Created a custom UI for engineers to simulate train runs (playback tool) and included functionality for accurate data to be read. (C# / XAML)
- Automated setup processes, optimizing log parsing and tool runtime efficiency to make the tool over 20x faster. (C#)

The University of Iowa, Undergraduate Teaching Assistant *January 2020 - May 2022*

- Guided students in gaining an understanding of the basics of coding (C++)
- Graded homeworks, held office hours, proctored exams, and aided a class of over 300 students learning to code

Education

Penn State University *December 2024*

Master of Science in Software Engineering **GPA: 3.8**

The University of Iowa *May 2022*

Bachelor of Science in Computer Science Engineering **GPA: 3.7 (Presidents List, Old Gold Scholar)**

Activities/Projects

The University of Iowa SpaceX and Hyperloop Team, Software Lead *September 2018 - May 2022*

- Worked in specific software group to develop and code certain aspects of hyperloop pod (C++)

National Voting System Application (Hackathon top 5 winner) *October 2020*

- Developed a full-stack national voting app, featuring custom ballots, real-time voting, and result tracking. (Java / SQL)

Skills

C++ • Java • Python • JavaScript (HTML, CSS) • C# • Azure • AWS • PostgreSQL • Git
CI/CD • Jenkins • Agile • Web App Development • Kubernetes • Terraform • React • Docker